

cvbets: an AI horse-race for CS2 predictions

cvbets is a simulated prediction tracker for professional Counter-Strike 2 matches. Several AI 'tipsters' (we call them agents) predict who will win each upcoming match. Every night, the system checks real results and keeps score. Think of it as a season-long competition between tipsters, refereed by software nobody can argue with. Important: no real money anywhere — this is bragging rights and statistics only.

The daily loop (runs itself at midnight)

Every night at 12:00am, an automated job runs four steps in order and posts a summary to our Discord channel:

1. Ingest	Downloads today's upcoming CS2 matches and the last 60 days of results from PandaScore, a professional esports data provider.
2. Pick	Each agent that hasn't already spoken on a match locks in its prediction — with a confidence level — before the match starts. Picks are locked forever once made.
3. Grade	For every match that finished, each agent's pick is marked WIN or LOSS and given a Brier score — a fairness metric that punishes confidently wrong predictions more than humble ones.
4. Publish	The standings table, upcoming picks, and recent results are rebuilt into a public website and summarized on Discord.

The referees: three tipsters compete now, more can join anytime

Every agent answers the same question — who wins this match, and how sure are you? — but each thinks differently:

chalkbot	The favorite-chaser	Checks each team's win rate over the last 30 days in our own database and backs the hotter team. If it doesn't know enough about a team, it picks team 1 at coin-flip confidence. The baseline everyone must beat.
form	The recency scout	Only looks at each team's last 5 matches, whoever the opponent. A team on a 4-1 run beats a team slumping at 1-4, even if the season stats disagree.
h2h	The history buff	Only cares about the all-time record between these two exact teams. Rivalries and style matchups matter more than current streaks. No history? Coin flip.

New agents — including LLM-powered ones that read the data and reason in plain language — can join later by submitting predictions through the same gate. Nobody gets special treatment.

The rules that keep it honest

Append-only ledger	Once a prediction is written, it can never be edited or deleted — like writing in pen. The only allowed change is the referee stamping WIN or LOSS on it once, after the match. No rewriting history, no 'I meant to pick the other team'.
The gate	Agents cannot touch the database directly. Predictions go through a checkpoint that rejects anything malformed: wrong team name, confidence outside 50-100%, missing reasoning, more than 100 words, or a second prediction on the same match.
One pick per agent per match	Enforced at the gate and in the database. No hedging, no do-overs.
Timestamps everywhere	Every pick and every grade is timestamped, so the full story of who-knew-what-when is preserved.

How scoring works (the Brier score)

A tipster doesn't just say a team name — it attaches a confidence from 0.50 (coin flip) to 1.00 (certain). The Brier score measures how well calibrated those confidences are: $\text{score} = (\text{confidence} - \text{outcome})^2$, where a win counts as 1 and a loss as 0. Lower is better. Saying 80% and being right earns 0.04; saying 80% and being wrong costs 0.64. A perfect 0.00 means you were right every time with total confidence; 0.25 is what you earn by always guessing at exactly 50%. The point: agents can't win by being lucky and loud — they have to be right at the rate they claim.

Current standings

As of this printing (nightly grading at 12:00am keeps moving these numbers):

Agent	Graded	W-L	Win rate	Brier
chalkbot	44	26-18	59.1%	0.2500
form	1	0-1	0.0%	0.2500
h2h	1	0-1	0.0%	0.2500

Scale so far: 2,588 matches tracked (2,502 finished, 86 upcoming), 304 predictions logged, 46 graded, 258 awaiting their matches. The new agents have each graded only one pick so far — their real records build up over the coming nights. chalkbot's 0.25 Brier is exactly what coin-flip confidence produces; as it now posts confidences up to 0.86, its Brier will either fall (good calibration) or rise (overconfidence exposed). That's the experiment.

Where it lives

Public dashboard	https://sntvnce.github.io/cvbets/ — standings, upcoming picks, results
Source code	https://github.com/sntvnce/cvbets — ~1,600 lines of Python, no frameworks
Nightly report	Posted to the #guru-updates Discord channel after every midnight run
Data	A single SQLite database (ledger.db) — the append-only source of truth

What's next

Three upgrades on the bench: (1) LLM expert agents that read the ledger and argue their picks in plain language, (2) a 'chief' agent that learns to weight the experts by their proven calibration and emits its own consensus picks, and (3) bookmaker odds, which unlock the real endgame — comparing every agent not just on accuracy but on whether its confidence would have made money against the market.